

Self-Learning: Address Decoding Techniques & 8259 Cascading

MPCA Module 4

> **Definition:** Address Decoding Techniques define how system address lines are mapped to peripheral chip selects (Absolute vs Partial), while **8259 Cascading** expands interrupt handling capacity from 8 to 64 priority channels.

1. Absolute vs Partial Address Decoding

ABSOLUTE ADDRESS DECODING:

- On a specific address, only one device is enabled.

PARTIAL ADDRESS DECODING:

- On a specific address, multiple devices can be enabled.

Comparison Table:

Feature	Absolute Decoding	Partial Decoding
Number of devices enabled	One	Multiple

2. 8259 Cascaded Mode (Master-Slave Configuration)

To handle more than 8 interrupts, one **Master 8259** is connected to up to **8 Slave 8259s** via the 3-bit Cascade Bus (**CAS_0, CAS_1, CAS_2**).

Cascading Operation Flow:

- Slave 8259 receives interrupt on its IR line and signals Master 8259 via Master's IR pin.
- Master 8259 sends interrupt signal to the system.

3. Quick Recall Flow

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